Kinematic Modelling of Five-Bar Bipedal Locomotion: An Analytical Approach

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***Abstract*—This paper presents an analytical framework for the kinematic modeling of a five-bar bipedal robot, which is a home-made experimental prototype. The five-bar linkage introduces complex holonomic constraints and closed-loop dependencies that require precise mathematical formulation. We derive the forward and inverse kinematic equations using an analytical approach, providing a deterministic solution for the foot trajectory relative to the hip base. The study further establishes a closed-form mapping between joint space and task space, this research provides as a first approach for path tracking and control a Kalman filter for straight trajectories. The proposed model is validated through numerical simulations and laboratory experiments, characterizing the workspace and motion primitives of five-bar bipedal locomotion.**

***Index Terms*—Bipedal robot, five-bar linkage, kinematic-motion, Kalman-filter, Walking-gait**

I. Introduction

Modern robotics shifts from heavy industrial arms to agile, mobile agents, where these systems are more relevant than ever. While massive humanoid robots often grab headlines, five-bar bipedal systems represent efficiency and simplicity. Five-bar limb mechanisms are able to control a foot's position using servomotors located near the hip, consequently reducing swing mass effects, making lighter and more responsive limbs (home-made experimental prototype Fig. 1ab). Unlike wheeled robots, bipeds navigate complex obstacles such as curbs or stairs. Their closed-loop nature makes them more rigid and energy-efficient than traditional serial-link robots.

A purpose of this research is to provide a physics-based bridge between theoretical modeling and physical experimentation. This approach is tied to holonomic constraints and closed-loop dependencies, which are non trivial issues. A five-bar system is a parallel multi-bar linkage that offers high payload-to-weight ratio (Fig. 1c). The structure shares the mechanical load across multiple bars. Besides, performs high precision as metric errors in one joint do not necessarily becomes as so accumulative as they do in serial multi-legged robots. This kind of robotic platform (Fig.1) provides a way to understand its inverse kinematics, both the joints positioning and the whole body displacement. The implementation of a Kalman filter is a suitable technique for sensor fusion in order to deal with noisy sensor data due to mechanical vibrations in real-time. In principle, research into motion primitives allows global motion planning by characterizing the limbs' workspace, and use it to perform multiple robot's gaits suitable for different terrains. A main summarized contribution of this paper is to provide a deterministic kinematic solution that allows for walking path tracking. Moreover, dynamic balance by using the Kalman Filter to estimate the robot's state and prevent it from tipping over. Essentially, the five-bar bipedal robot is deployed to solve the problems of legged locomotion.

This manuscript is organized as follows: Section II provides a comparative analysis of related work. Section III introduces the robot's architecture and mechanical design. Section IV derives the forward and inverse kinematics of the five-bar linkage. Section V presents the algebraic deduction for the robot's multi-part kinematic structure. Section VI details the implementation of the Kalman filter for motion estimation. Finally, Section VII offers concluding remarks.

II. Related Work

The design and performance characterization of underactuated bipedal systems have been extensively documented, notably in [1], while more recent efforts have employed derivative-free optimization to refine leg parameters for ZMP estimation and energetic efficiency [2]. Within this domain, five-bar mechanisms have emerged as a versatile solution, facilitating bio-inspired motion [3], high-agility maneuvering [4], and the exploration of fundamental kinematic properties [5]. To address the inherent instabilities of legged locomotion, researchers have further implemented inverted pendulum control strategies to regulate tilt during high-speed gait [6]. Beyond traditional bipedalism, multi-bar linkages have found diverse applications—serving as sophisticated suspension systems for self-balancing platforms [7], [8] and as bio-mimetic propulsion for underwater robotics [9], [10]. Similarly, the Klann linkage continues to represent a foundational framework for quadrupedal locomotion across a wide spectrum of experimental and industrial paradigms [11]–[14].

III. Robot Structure

This section details the robotic platform design [15], encompassing its mechanical structure, computational architecture, instrumentation, and sensor-actuator integration. Figure 1ab illustrates the bipedal experimental prototype alongside its

CAD model. Additionally, figure 1c clarifies the limb's linkage arrangement, demonstrating its geometric consistency with the isolated five-bar mechanism shown in Fig. 3.

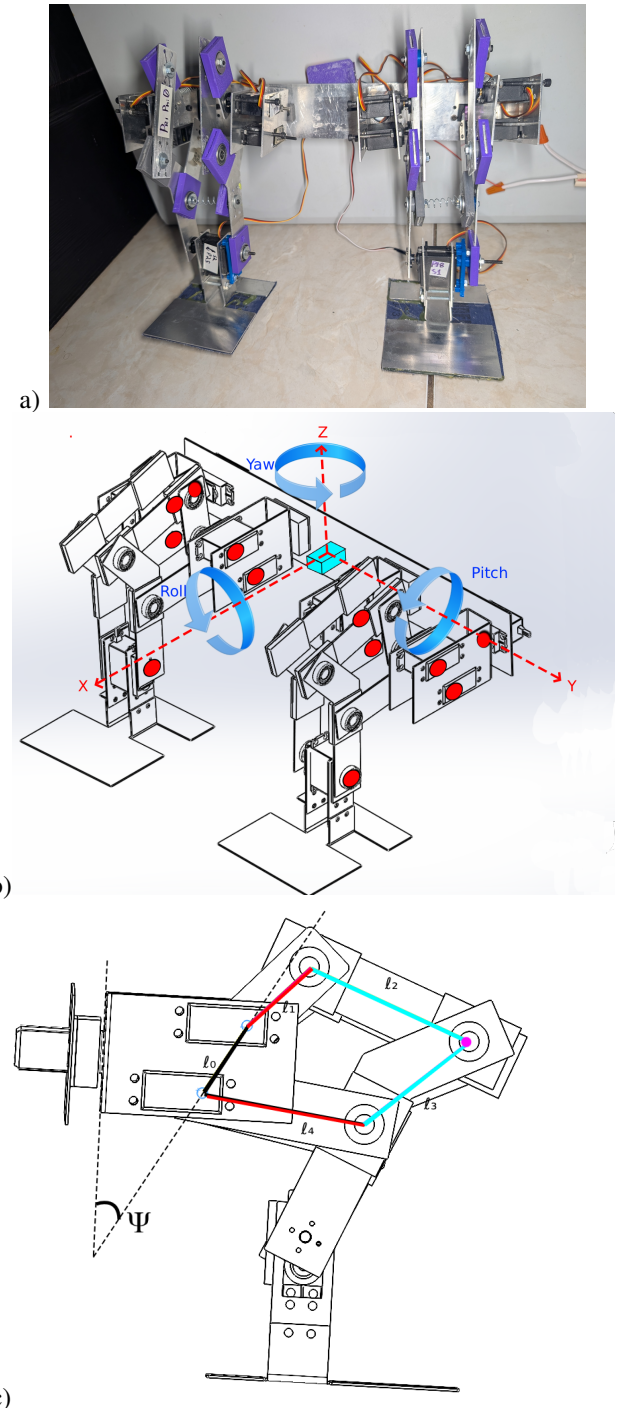


Fig. 1. Robotic bipedal five-bar linkage limbs. a) Experimental prototype. b) Detailed CAD model showing motor-potentiometer placement (red points) and central IMU (blue rectangle). c) Limb's side view.

The robotic architecture, shown in Figure 2, comprises an Orange Pi Zero (Debian Linux) interfacing with an ESP32 slave module for low-level control of twelve servomotors. The instrumentation includes a 9-axis IMU located at the center of mass and twelve potentiometers used as angular transducers to monitor the roll and pitch of each limb.

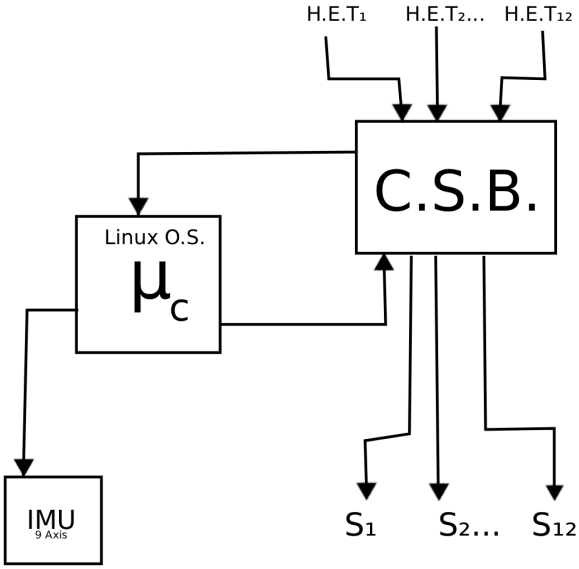


Fig. 2. The master unit (μ_c) interfaces with the Control and Sensing Board (C.S.B.) to manage an array of twelve Hall Effect Transducers (H.E.T. $_1$ to H.E.T.$_{12}$) and twelve servos (S_1 to S_{12}).

IV. Limb's Kinematics

A. Forward Kinematics

The kinematics development of a Five-Bar Bipedal locomotion is based on the analysis of one closed-chain 5-bar mechanism. This analysis will be divided into 2 main stages, first the forward kinematics (Fig.3a), which allows us to determine the position of our end-effector from the actuator angles, and the inverse kinematics will allow us to determine the angles of our actuators from desired coordinates for our end-effector (Fig.3b), alternative approach in [16]. We begin by defining our X and Y coordinates from the end of ℓ_1 and ℓ_4 with our following expressions:

$$x_1 = \ell_1 \cos(\theta_1); \quad y_1 = \ell_1 \sin(\theta_1); \tag{1}$$

$$x_4 = \ell_0 + \ell_4 \cos(\theta_4); \quad y_4 = \ell_4 \sin(\theta_4). \tag{2}$$

These positions are necessary to know the instant distance from the ends of ℓ_1 and ℓ_4 which we call ℓ_e and is defined geometrically by:

$$\ell_e = \sqrt{(x_1 - x_4)^2 + (y_1 - y_4)^2} \tag{3}$$

ℓ_e is a measurement from the inter-geometric relation between the angles θ_1 and θ_4, if the following condition is true;

$$(\ell_2 + \ell_3) < \ell_e \tag{4}$$

Then, it is known that the closed kinematic chain cannot reach the desired posture. Thus, to determine inner angles β, α and γ, the sine and cosine rules obtained the expressions:

$$\beta = \arccos\left(\frac{{\ell_e}^2 - {\ell_2}^2 - {\ell_3}^2}{-2\ell_2\ell_3}\right), \tag{5}$$

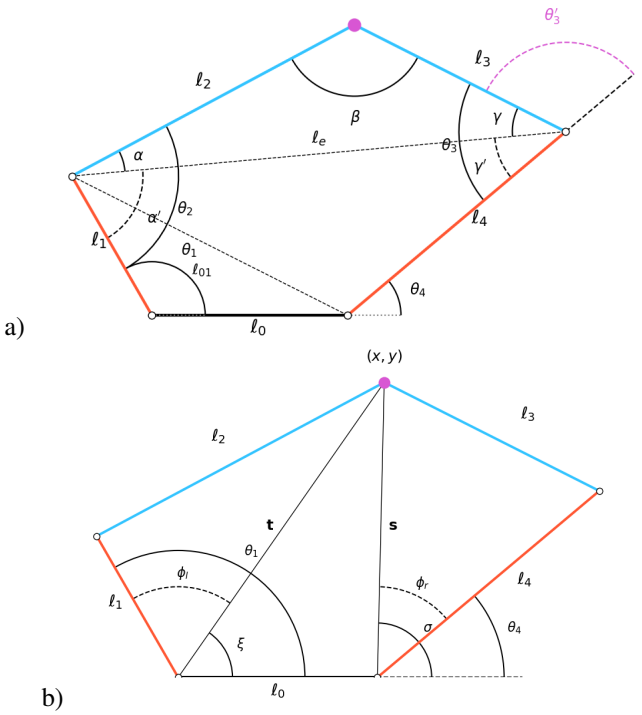


Fig. 3. Limb's 5-bar linkage geometric model. a) Forward kinematics. b) Backward kinematics.

$$\alpha = \arcsin(\frac{\ell_3}{\ell_e}\sin(\beta)), \tag{6}$$

$$\gamma = \arcsin(\frac{\ell_2}{\ell_e}\sin(\beta)). \tag{7}$$

To continue we require the local version of the γ, which we'll call γ'. However, to calculate γ' we must first determine the instantaneous distance from the top end of ℓ_1 and the right end of ℓ_0 as seen on Figure 3. By applying the law of cosines, we obtain the expression for this distance, ℓ_{01}:

$$\ell_{01} = \sqrt{{\ell_0}^2 + {\ell_1}^2 - 2\ell_1\ell_0\cos(\theta_1)}, \tag{8}$$

which we use to calculate γ' using the expression:

$$\gamma' = \arccos\left(\frac{{\ell_{01}}^2 - {\ell_4}^2 - {\ell_e}^2}{-2\ell_e\ell_4}\right) \tag{9}$$

As observed in geometric scheme in Fig.3, global angle θ_3 and local angle θ_3' are supplementary, as they lie on the same horizontal axis. Hence, their relationship is defined as:

$$\theta_3' = \pi - \theta_3, \tag{10}$$

which is needed to calculate the coordinates of our end-effector using:

$$p_x = \ell_0 + \ell_4\cos(\theta_4) + \ell_3\cos(\theta_4 + {\theta_3}'), \tag{11}$$

$$p_y = \ell_4\sin(\theta_4) + \ell_3\sin(\theta_4 + {\theta_3}'). \tag{12}$$

B. Inverse Kinematics

Conversely, inverse kinematics aims to determine the active joint angles, θ_1 and θ_4 required to position our end-effector at known Cartesian coordinates. By defining the position $\mathbf{t}$, pointing out to the end-effector, its magnitude and true elevation angle relative to the local horizontal ground plane are determined.

$$||\mathbf{t}|| = \sqrt{p_x^2 + p_y^2}, \tag{13}$$

$$\xi = \arctan\left(\frac{p_y}{p_x}\right). \tag{14}$$

Thus, to find the angle between ℓ_1 and $\mathbf{t}$, the cosine rule yields:

$$\phi_l = \arccos\left(\frac{\ell_2^2 - ||\mathbf{t}||^2 - \ell_1^2}{-2\ell_1||\mathbf{t}||}\right) \tag{15}$$

And the angular position of the left actuator is defined by:

$$\theta_1 = \xi + \phi_l \tag{16}$$

Let us continue the analysis by getting the angular position of the right actuator, which is calculated by defining a positional vector defined as $\mathbf{s}$ and its magnitude and elevation with respect to the ground come from the equations:

$$||\mathbf{s}|| = \sqrt{(p_x - \ell_0)^2 + p_y^2} \tag{17}$$

$$\sigma = \arctan\left(\frac{p_y}{p_x - \ell_0}\right) \tag{18}$$

The angle between ℓ_4 and $\mathbf{s}$ is a cosine rule such that,

$$\phi_r = \arccos\left(\frac{\ell_3^2 - ||\mathbf{s}||^2 - \ell_4^2}{-2||\mathbf{s}||\ell_4}\right). \tag{19}$$

Due to the geometric configuration of the right side, the internal angle is subtracted from σ, so the angular position of the right actuator is defined by:

$$\theta_4 = \sigma - \phi_r \tag{20}$$

According to Fig. 4, it is a track of a square-shape trajectory to validate the inverse kinematics accuracy, demonstrating precise path tracking and correct execution of sharp directional changes.

V. Biped's Kinematics

Due to the fact that the legs of the platform do not operate on a static plane, it is necessary to transform the local system coordinates to a global 3D reference system, with its origin located on the center of its hip. This transformation consists of 3 fundamental rotations, a translation defined by:

$$\mathbf{R_z}\left(-\frac{\pi}{2}\right) = \begin{pmatrix} \cos\left(\frac{\pi}{2}\right) & -\sin\left(-\frac{\pi}{2}\right) & 0 \\ \sin\left(-\frac{\pi}{2}\right) & \cos\left(-\frac{\pi}{2}\right) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{21}$$

$$\mathbf{R_y}\left(\frac{\pi}{2}\right) = \begin{pmatrix} \cos\left(\frac{\pi}{2}\right) & 0 & \sin\left(\frac{\pi}{2}\right) \\ 0 & 1 & 0 \\ -\sin\left(\frac{\pi}{2}\right) & 0 & \cos\left(\frac{\pi}{2}\right) \end{pmatrix}. \tag{22}$$

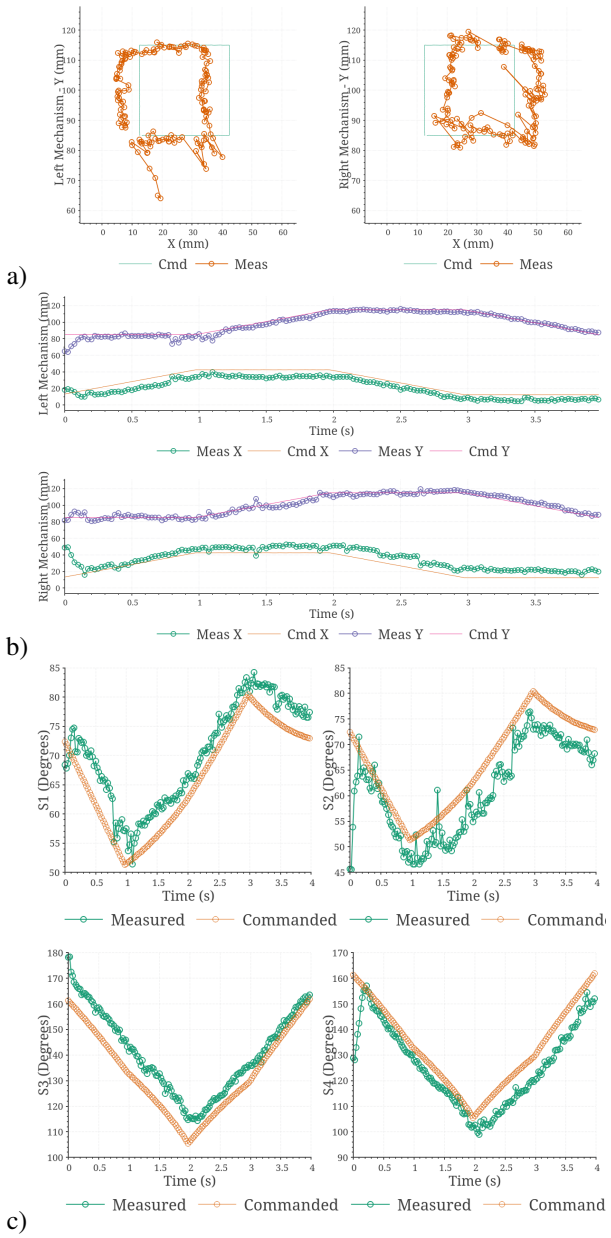


Fig. 4. Limb kinematics validation for a square trajectory. a) Spatial workspace tracking in task space. b) End-effector displacement over time. c) Active joints angular commands and measurements.

The leg rotation is the angle between ℓ_0 and the hip, set as ψ,

$$\mathbf{R_z}(\psi) = \begin{pmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{23}$$

The final rotation matrix is multiplying the individual rotation matrices, giving us the expression:

$$\mathbf{R_{final}} = \mathbf{R_z}\left(-\frac{\pi}{2}\right)\mathbf{R_y}\left(\frac{\pi}{2}\right)\mathbf{R_z}(\psi) \tag{24}$$

Finally, the three-dimensional position $\mathbf{p_{3D}}$ is given by applying the rotation to the local vector and adding the translation vector, which is obtained from the offset on the y axis;

$$\mathbf{p_{3D}} = \mathbf{R_{final}}\mathbf{p_{local}} + \begin{pmatrix} 0 \\ y_{offset} \\ 0 \end{pmatrix} \tag{25}$$

Now knowing the limbs and biped's kinematics we shall continue with the workspace of both. The limbs workspace is evaluated sweeping both actuators angles θ_1 and θ_4 across their physical limits $[0, \pi]$ while also filtering out unreachable coordinates by enforcing the geometric constraint indicated on Equation 4 Subsequently, the 3D workspace of the bipedal

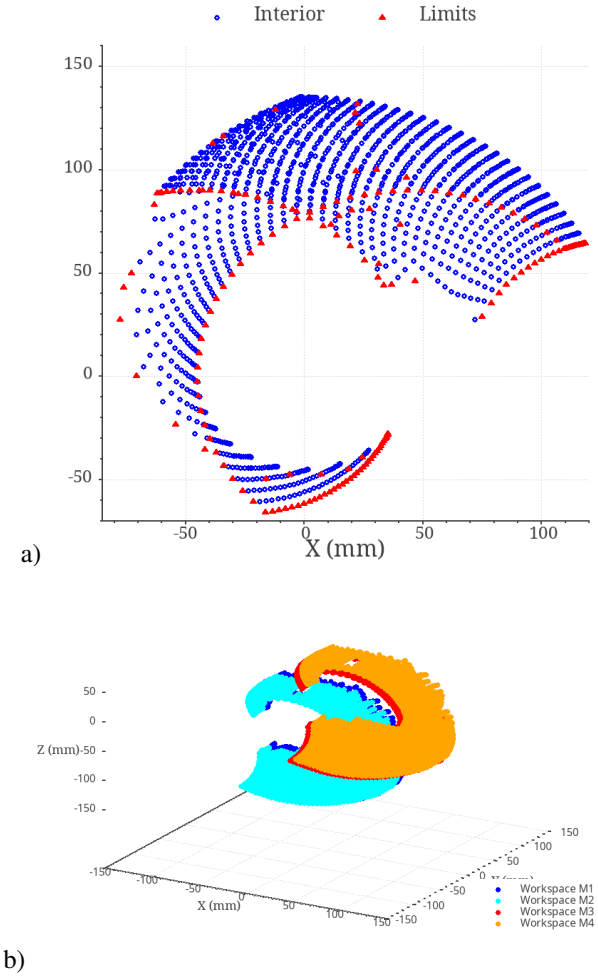


Fig. 5. Bipedal gaits workspace. a) Planar; and b $_{c)}$ volumetric.

platform is computed by mapping the 2D boundaries into the spatial domain. Applying the rotation and offset of each leg to the planar workspace we add an extra rotation which will allow the legs to do abduction and adduction movements with limits of $[-15\frac{\pi}{180}, 15\frac{\pi}{180}]$ and doing a sweep of all the active articulations giving us a comprehensive mapping of the biped's total reachable volume, which is crucial for trajectory planning and gait generation. The bipedal locomotion is achieved by driving the active joints with harmonic continuous functions.

Let $\phi_1(t)$ and $\phi_4(t)$ be the angular positions for the primary and secondary actuators of the right leg, defined as:

$$\phi_{1_R}(t) = \phi_{1_{base}} + A\sin(\omega t),$$

$$\phi_{4_R}(t) = \phi_{4_{base}} + A\sin\left(\omega t + \frac{\pi}{2}\right),$$

where ϕ_{base} denotes the central angular offset, A is the amplitude, and ω the angular frequency of the gait cycle. To achieve an alternating walking pattern, the left leg actuators operate with a phase shift of π radians relative to the right leg, ensuring that while one leg is in the swing phase, the other remains in the stance phase:

$$\phi_{1_L}(t) = \phi_{1_{base}} + A\sin(\omega t + \pi),$$

$$\phi_{4_L}(t) = \phi_{4_{base}} + A\sin\left(\omega t + \frac{3\pi}{2}\right).$$

Finally, the 3D spatial trajectory of each foot, $\mathbf{P}_{foot}(t)$, is computed by mapping these time-varying joint angles through the forward kinematics function $\mathbf{FK}$ and applying the hip's rotational matrix $\mathbf{R}_{final}$ and translation vector $\mathbf{T}_{offset}$:

$$\mathbf{P}_{foot}(t) = \mathbf{R}_{final} \cdot \mathbf{FK}(\phi_1(t), \phi_4(t)) + \mathbf{T}_{offset}.$$

Figure 6b shows both limbs synchronized gaits, and such cycloids are ones of the so numerous existing in the set of the workspace (Fig.5). Figure 6a includes the input angles in active joints to yield the cycloid.

VI. Motion State Estimation

In a first approach, to ensure stable locomotion along straight trajectories, accurate estimation of the robot's position and speed is implemented. Sensor data inherently contain noise, which can destabilize the gait cycle. To reduce this, a Kalman filter was implemented for path tracking. The motion of the biped is modeled as a linear dynamic system. The state vector $\mathbf{x_k} \in \mathbb{R}^4$ at time step k contains the 2D position and velocity of the robot, defined as $\mathbf{x_k} = [x, y, \dot{x}, \dot{y}]^T$. The system is governed by the state transition equation that is the result of discretizations of forward Euler:

$$\mathbf{x_{k+1}} = \mathbf{A}\mathbf{x_k} + \mathbf{B}\mathbf{u} + \mathbf{w_k}. \tag{26}$$

Based on the constant velocity model, the state transition matrix $\mathbf{A}$ and the control input matrix $\mathbf{B}$ are defined by the time step Δt:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} \frac{\Delta t^2}{2} & 0 \\ 0 & \frac{\Delta t^2}{2} \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}. \tag{27}$$

The control vector is given by the acceleration $\mathbf{u} = [\ddot{x}, \ddot{y}]^\top$ and $\mathbf{w_k} \sim \mathcal{N}(0, \mathbf{Q})$ represents the process perturbation vector with covariance matrix $\mathbf{Q}$. The process noise covariance matrix was empirically defined as $\mathbf{Q} = \text{diag}(0.05^2, 0.05^2, 0.01^2, 0.001^2)$ to account for unmodeled physical disturbances during locomotion, such as mechanical backlash in the servomotors. The

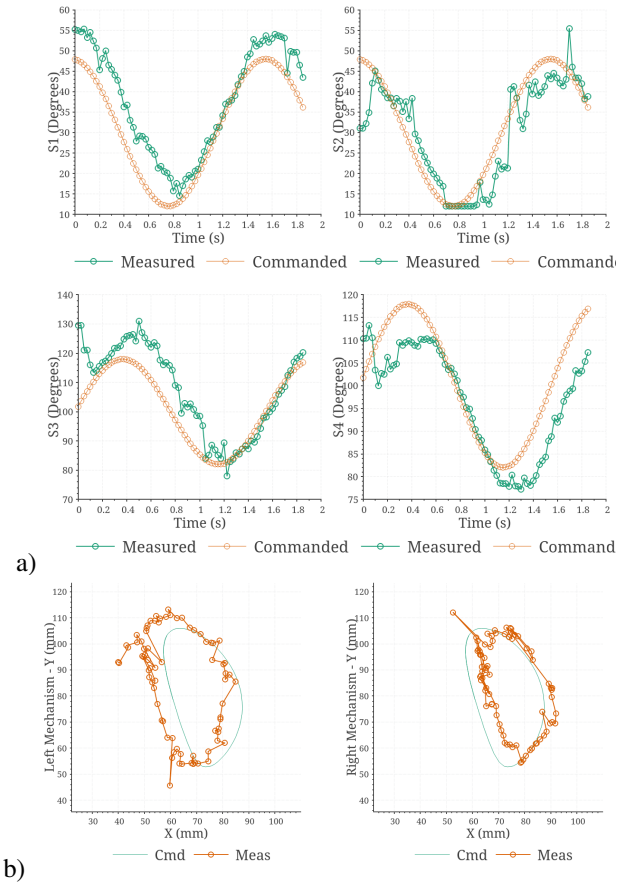


Fig. 6. Cycloidal gait. (a) Angular trajectories of actuators over time. (b) Task space, comparing commanded cycloidal paths versus measured trajectory

measurement vector $\mathbf{z_k} \in \mathbb{R}^2$, containing the noisy positional sensor readings, is modeled as:

$$\mathbf{z_k} = \mathbf{H}\mathbf{x_k} + \mathbf{v_k} \tag{28}$$

Where the observation matrix $\mathbf{H}$ maps the true state space into the observed space, tracking only the x and y coordinates:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \tag{29}$$

The measurement noise $\mathbf{v_k}$ is bounded by the sensor's covariance matrix $\mathbf{R}$. Based on the system's experimental characterization, the noise standard deviation for the measurements was established at $\sigma = 0.01$, defining the observation covariance as $\mathbf{R} = \text{diag}(0.01^2, 0.01^2)$. The Kalman filter operates recursively. First, the prediction step projects the current state and error covariance matrix $\mathbf{P_k}$ forward:

$$\hat{\mathbf{x}}_{k+1} = \mathbf{A}\hat{\mathbf{x}}_k + \mathbf{B}\mathbf{u}, \tag{30}$$

$$\mathbf{P}_{k+1} = \mathbf{A}\mathbf{P}_k\mathbf{A}^T + \mathbf{Q}. \tag{31}$$

Thus, the update step corrects the estimate using a new measurement $\mathbf{z_{k+1}}$ by calculating the Kalman Gain $\mathbf{K_{k+1}}$:

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1}\mathbf{H}^T(\mathbf{H}\mathbf{P}_{k+1}\mathbf{H}^T + \mathbf{R})^{-1} \tag{32}$$

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{k+1} + \mathbf{K}_{k+1}(\mathbf{z}_{k+1} - \mathbf{H}\hat{\mathbf{x}}_{k+1}) \quad (33)$$

$$\mathbf{P}_{k+1} = (\mathbf{I} - \mathbf{K}_{k+1}\mathbf{H})\mathbf{P}_{k+1} \quad (34)$$

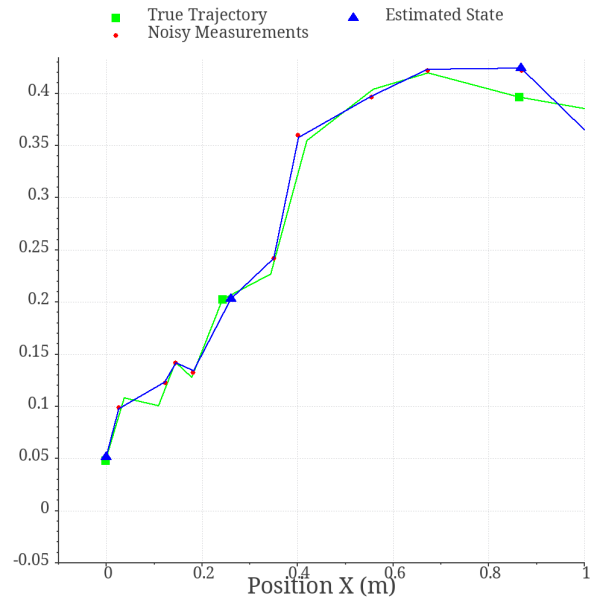


Fig. 7. Motion state estimation using a discrete-time Kalman filter.

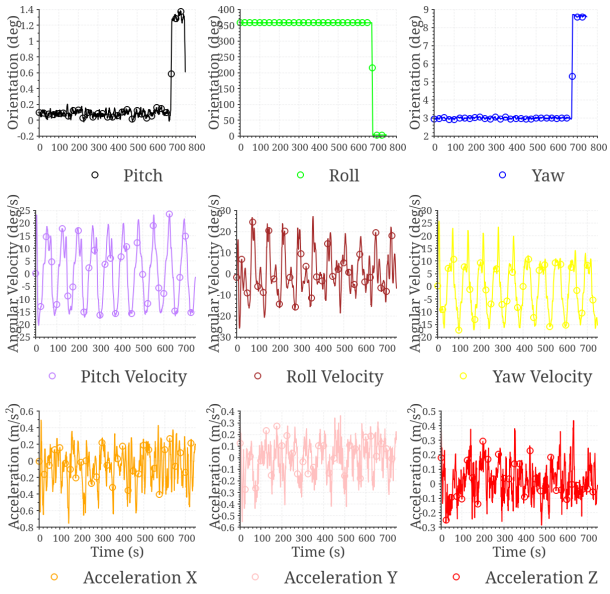


Fig. 8. Raw IMU data from walking experiment.

VII. Conclusion

In this study, we successfully developed a rigorous analytical framework for the kinematics of an experimental five-bar bipedal prototype. By establishing a deterministic, closed-form mapping between joint and task spaces, we eliminated the computational overhead typically associated with iterative numerical solvers. The alignment between our simulations and experimental results although it is still undergoing, but it is found that the proposed model provides a reliable foundation for real-time trajectory tracking and precise foot placement in closed-loop robotic systems. Future work involves the inclusion of dynamic control and experimental multi-gait walking. Furthermore, an upgrade towards the Unscented Kalman filter to combine IMU data and the limbs angular transducers.

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References

[1] K. G. Gim, J. Kim and K. Yamane, “Design and Fabrication of a Bipedal Robot Using Serial-Parallel Hybrid Leg Mechanism,” 2018 IEEE/RSJ Intl. Conf. on Intel Robot and Sys (IROS), Madrid, Spain, 2018, pp. 5095-5100.

[2] K. Keawhanam, R. Chuengpichanwanich, C. Khlowutthiwat, R. Chaichaowarat, “Optimal Five-bar Legged Design for Energy-Efficient Bipedal Robot,” 2023 IEEE Intl. Conf. on Robotics and Biomimetics (ROBIO), Koh Samui, Thailand, 2023, pp. 1-6.

[3] Z. Wang, W. Ge, Y. Zhang, B. Liu, B. Liu, S. Jin, Y. Li, “Optimization Design and Performance Analysis of a Bionic Knee Joint Based on the Geared Five-Bar Mechanism.” Bioengineering 2023, 10, 582.

[4] C. Fisher, A. Blom, and A. Patel, “Baleka: A Bipedal Robot for Studying Rapid Maneuverability,” Frontiers in Mech Eng, vol. 6, p. 54, 2020.

[5] H. Shin, T. Ishikawa, T. Kamioka, K. Hosoda, and T. Yoshiike, “Mechanistic Properties of Five-bar Parallel Mechanism for Leg Structure Based on Spring Loaded Inverted Pendulum,” in 2019 IEEE-RAS 19th Int. Conf. on Humanoid Robots (Humanoids), pp. 320–327, 2019.

[6] H. Gao et al., “Stable Rapid Sagittal Walking Control for Bipedal Robot Using Passive Tendon,” Actuators, vol. 13, no. 7, p. 240, 2024.

[7] S.J. Favela, E.A. Martínez-García, “Rolling Biped Polynomial Motion Planning”, Motion Planning, pp. 1-22, IntechOpen, 2022

[8] N. Mao, J. Chen, E. Spyrakos-Papastavridis, and J. S. Dai, “Dynamic modeling of wheeled biped robot and controller design for reducing chassis tilt angle,” Robotica, vol. 42, no. 8, pp. 2713–2741, 2024.

[9] M. Montoya, R. Torres-Córdoba, E. Magid, E.A. Martínez-García, “Electromyography-Based Biomechanical Cybernetic Control of a Robotic Fish Avatar”, Machines, 2024, 12, 124, p. 1-41, MDPI

[10] E. A. Martinez-Garcia, R. Lavrenov, E. Magid, “Robot fish caudal propulsive mechanisms: a mini-review”, AI, Computer Science and Robotics Technology, p. 1-27, IntechOpen, June 2022

[11] J. A. Aguilera Jímenez, E. A. Martínez-García, “Dynamic Modelling and Control of an Underactuated Quasi-Omnidireccional Hexapod,” Handbook of Research on Advanced Mechatronic Systems and Intelligent Robotics, pp.377-400, 2019, IGI Global, USA

[12] V.S. Rajashekhar, D. Ghose, A.S.A. Doss, “Optimal Kinematic Design of a Robotic Lizard Using Four-Bar and Five-Bar Mechanisms.” In: Kumar, R.S., Sanyal, S., Pathak, P.M. (eds) Recent Advances in Machines, Mechanisms, Materials and Design. iNaCoMM 2023. Lecture Notes in Mechanical Engineering. Springer, Singapore.

[13] J.K. Sheba, R.E. Mohan, E.A. Martínez-García, L. Tan-Phuc, “Trajectory Generation and Stability Analysis for Reconfigurable Klann Mechanism based Walking Robot”, Robotics, 5(3),13; pp. 1-20, mdpi, 2016

[14] J. Kulandaidaasan, R.E. Mohan, E.A. Martínez-García, Le Tan-Phuc, “Synthesizing reconfigurable foot traces using a Klann mechanism”, Robotica, Cambridge University Press, Vol.33, Mar 2015.

[15] V. Batto, T. Flayols, N. Mansard and M. Vulliez, “Comparative Metrics of Advanced Serial/Parallel Biped Design and Characterization of the Main Contemporary Architectures,” 2023 IEEE-RAS 22nd Intl. Conf. on Humanoid Robots (Humanoids), Austin, TX, USA, 2023, pp. 1-7.

[16] R. C. Liu, G. Y. Ma, Y. Chen, S. Han, J. Gao, “Four-legged robot design and gait planning”, J. of Physics: Conf. Series, 1087(6), 2018.